# Supporting Information for: Hard Shape Constraints versus Penalties for Temperature Extrapolation of MOF and COF Water Isotherms

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Contents: Tables S1–S9; Figure S1; Notes S1–S4 (synthetic worlds and power; synthetic truths for the isosteric heat; digitisation error; model definitions); references.

## Table S1: Data

| Material | Class | Set | Data | Source DOI | $T$ range (K) | $n_T$ | Points per $T$ | $p/p_{\mathrm{sat}}$ range | $q_{\mathrm{ref}}$ (mol kg$^{-1}$) |
|---|---|---|---|---|---|---|---|---|---|
| COF-432 | COF | conf. | vector | 10.1021/jacs.9b13094 | 283.15–313.15 | 3 | 28–29 | 0.063–0.870 | 16.04 |
| COF-482-hydrazide | COF | conf. | raster | 10.1021/acscentsci.2c00398 | 288.15–308.15 | 3 | 25–51 | 0.08–0.898 | 34.70 |
| COF-$SO_3$H | COF | conf. | vector | 10.1002/cssc.202301906 | 298.15–318.15 | 3 | 18–18 | 0.0003–0.900 | 17.04 |
| HFPTP-DMePDA-COF | COF | conf. | raster | 10.1038/s41467-021-27128-4 | 283.15–298.15 | 3 | 9–18 | 0.39–0.897 | 7.91 |
| HFPTP-PDA-COF | COF | conf. | raster | 10.1038/s41467-021-27128-4 | 283.15–298.15 | 3 | 24–32 | 0.0041–0.898 | 16.78 |
| NO-PI-3-COF | COF | conf. | vector | 10.1021/jacs.3c02572 | 288.15–308.15 | 3 | 30–37 | 0.0091–0.882 | 12.74 |
| PI-3-COF | COF | conf. | vector | 10.1021/jacs.3c02572 | 288.15–308.15 | 3 | 28–37 | 0.01–0.887 | 24.80 |
| TFBCz-PDA-COF | COF | conf. | raster | 10.1038/s41467-021-27128-4 | 283.15–298.15 | 3 | 29–30 | 0.0023–0.899 | 27.26 |
| Al-fumarate | MOF | conf. | raster | 10.1021/acscentsci.9b00745 | 283.15–323.15 | 3 | 56–59 | 0.049–0.892 | 23.86 |
| CALF-25 | MOF | conf. | ISODB | 10.1021/ja306812r | 298.00–313.00 | 3 | 17–20 | 0.018–0.894 | 3.75 |
| $Co_2Cl_2$(BTDD) | MOF | conf. | vector | 10.1021/acscentsci.7b00186 | 283.00–298.00 | 3 | 31–34 | 0.0072–0.862 | 53.97 |
| Mg-SU-102 | MOF | conf. | numeric | 10.1038/s41467-025-59551-2 | 288.15–328.15 | 9 | 21–21 | 0.0013–0.819 | 21.11 |
| $Zn_2$(bptc) | MOF | conf. | ISODB | 10.1021/ja060946u | 288.00–318.00 | 4 | 25–33 | 0.023–0.885 | 8.38 |
| TFPPy-PDA-COF | COF | dev. | raster | 10.1038/s41467-021-27128-4 | 283.15–298.15 | 3 | 7–15 | 0.46–0.900 | 35.41 |
| CuBTC | MOF | dev. | ISODB | 10.1016/j.ijheatmasstransfer.2012.07.068 | 293.00–358.00 | 9 | 23–41 | 0.0029–0.508 | 25.29 |
| MIL-100 | MOF | dev. | ISODB | 10.1016/j.ijheatmasstransfer.2012.07.068 | 293.00–358.00 | 9 | 25–35 | 0.011–0.603 | 18.11 |
| MIL-160 | MOF | dev. | ISODB | 10.1002/adma.201502418 | 293.00–313.00 | 3 | 32–54 | 0.0052–0.884 | 21.79 |

Table S1. The 13 confirmatory materials (conf.) and the four development materials (dev.) after the inclusion rules and the preprocessing. Data: how the points were obtained (vector or raster digitisation of the published figure, numeric source data, or the NIST adsorption database, ISODB). $n_T$: number of temperatures; points per temperature with $p/p_{\mathrm{sat}} \leq 0.90$; $q_{\mathrm{ref}}$: the largest uptake over all temperatures in this range, used only for scoring. Every decision and its reason is in data/inclusion_log.csv.

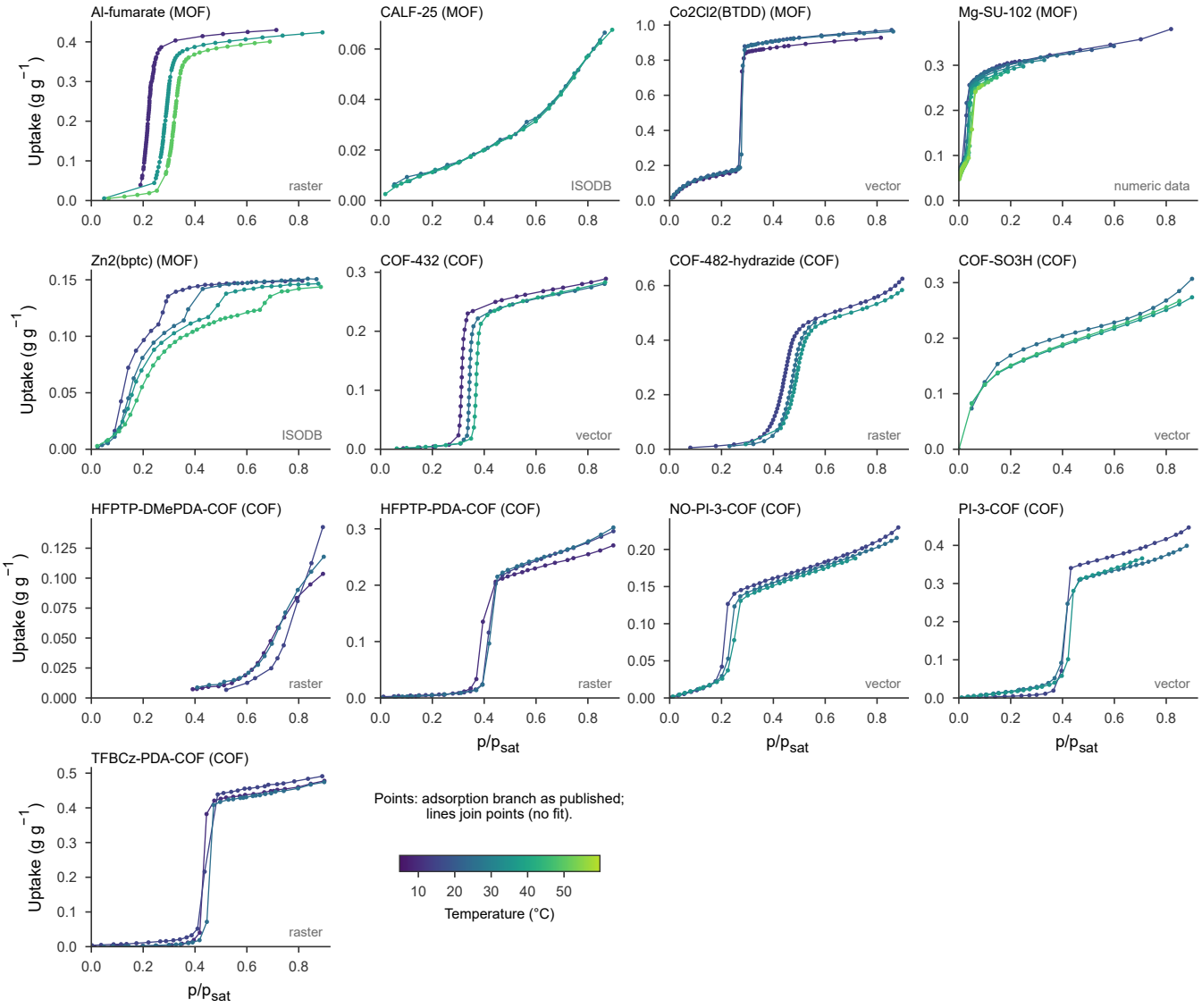



Figure S1. The confirmatory data: the adsorption isotherms of the 13 materials at every temperature, as used (relative pressure up to 0.90), coloured by temperature; points as published, joined by thin lines with no fit. The label in each panel gives how the points were obtained.

## Tables S2 to S7: Results in detail

| | | | $R_m = E_P/E_X$ | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Material | Class | $n_{train}$ | H1b | V-pen | H1c | A-hard | H1d | F-none | H1e | C1 | H1a | F-hard† | V-none† | F-rel† |
| COF-432 | COF | 2 | 0.86 | 0.83 | 0.37 | 0.98 | 0.70 | 0.84 | 1.58 | | | | | |
| COF-482-hydrazide | COF | 2 | 0.95 | 1.18 | 0.35 | 1.03 | 0.22 | 0.96 | 0.72 | | | | | |
| COF-$SO_3$H | COF | 2 | 10.91 | 9.92 | 6.19 | 8.58 | 6.04 | 9.80 | 5.35 | | | | | |
| HFPTP-DMePDA-COF | COF | 2 | 1.23 | 4.60 | 0.51 | 0.74 | 1.14 | 0.94 | 0.95 | | | | | |
| HFPTP-PDA-COF | COF | 2 | 0.99 | 2.28 | 1.01 | 1.01 | 0.85 | 1.00 | 0.95 | | | | | |
| NO-PI-3-COF | COF | 2 | 2.47 | 1.55 | 1.07 | 2.47 | 0.20 | 2.09 | 2.69 | | | | | |
| PI-3-COF | COF | 2 | 1.10 | 4.81 | 1.50 | 2.20 | 0.22 | 1.45 | 1.99 | | | | | |
| TFBCz-PDA-COF | COF | 2 | 0.78 | 3.14 | 0.99 | 0.97 | 0.39 | 0.94 | 0.83 | | | | | |
| Al-fumarate | MOF | 2 | 0.87 | 0.49 | 0.28 | 0.90 | 0.39 | 0.94 | 1.19 | | | | | |
| CALF-25 | MOF | 2 | 0.94 | 0.98 | 0.39 | 0.59 | 0.77 | 0.90 | 0.41 | | | | | |
| $Co_2Cl_2$(BTDD) | MOF | 2 | 1.44 | 1.13 | 1.02 | 0.92 | 0.59 | 1.38 | 1.92 | | | | | |
| Mg-SU-102 | MOF | 8 | 1.01 | 0.73 | 0.24 | 1.56 | 0.28 | 1.07 | 0.31 | | | | | |
| $Zn_2$(bptc) | MOF | 3 | 1.27 | 0.32 | 0.66 | 1.66 | 0.49 | 1.34 | 0.67 | | | | | |
| Hodges-Lehmann $R$ | | | 1.10 | 1.51 | 0.63 | 1.23 | 0.49 | 1.14 | 1.13 | | | | | |

Table S2. Error ratio of P against each comparator per material, $R_m = 10^{d_m}$, with $d_m$ the difference of the mean of $\log_{10} E$ over E-high and E-low; below one, P extrapolated better. $n_{train}$: the smallest number of training temperatures over the two tasks. The last row is the Hodges-Lehmann estimate over the 13 materials (main text, Figure 2a). Columns marked † (H1a, V-none, F-rel) are descriptive comparisons.

| | | Shape violations (H3) | | | | | | Isosteric heat (fits with any point) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | Fits | In domain | Wilson 95 % | Band | Strict | All tasks | Class | Undefined | Outside 0–150 | $< 0$ | $> 150$ |
| P | 26 | 0 | 0.00–0.13 | 0 | 0 | 0/39 | none | 0 | 14 | 10 | 6 |
| F-hard | 26 | 0 | 0.00–0.13 | 0 | 0 | 0/39 | none | 0 | 23 | 20 | 16 |
| A-hard | 26 | 0 | 0.00–0.13 | 0 | 0 | 0/39 | none | 0 | 0 | 0 | 0 |
| C1 | 26 | 0 | 0.00–0.13 | 0 | 0 | 0/39 | none | 0 | 2 | 1 | 1 |
| P-K | 2 | 0 | 0.00–0.66 | 0 | 0 | 0/3 | none | 0 | 2 | – | – |
| V-pen | 26 | 9 | 0.19–0.54 | 10 | 11 | 12/39 | some | 11 | 12 | 12 | 5 |
| V-none | 26 | 8 | 0.17–0.50 | 9 | 10 | 12/39 | some | 8 | 15 | 11 | 8 |
| F-pen | 26 | 13 | 0.32–0.68 | 12 | 24 | 19/39 | often | 23 | 26 | – | – |
| A-pen | 26 | 16 | 0.43–0.78 | 16 | 21 | 25/39 | often | 16 | 0 | – | – |
| A-none | 26 | 15 | 0.39–0.74 | 15 | 19 | 24/39 | often | 17 | 0 | – | – |
| F-rel | 26 | 22 | 0.66–0.94 | 22 | 26 | 34/39 | often | 25 | 26 | – | – |
| F-none | 26 | 26 | 0.87–1.00 | 26 | 26 | 39/39 | often | 25 | 26 | 24 | 26 |

Table S3. Violating fits per model over the 26 fits of E-high and E-low. Shape violations: In domain, the pre-registered H3 count, a fall of more than 2% of $q_{\mathrm{ref}}$ below the running maximum or an uptake below $-2\%$ of $q_{\mathrm{ref}}$, inside the penalty domain; Wilson 95% interval of its rate; Band, the same rule on the rest of the grid, up to 20 K beyond the measured temperatures and to 0.95 $p_{\mathrm{sat}}$; Strict, any decrease of uptake with pressure or any negative uptake anywhere on the grid, with no tolerance; All tasks, the in-domain count adding I-mid; Class, the pre-registered word. Isosteric heat (pre-registered strict counts, omitted from the first draft; ledger X-5): fits with any grid point where the heat is undefined, or outside 0 to 150 $\mathrm{kJ\,mol^{-1}}$, from the run records; that count split into negative heats and heats above 150 $\mathrm{kJ\,mol^{-1}}$ from exact refits of the reported seeds (ledger X-4; computed for the models in the H1 comparisons and V-none, – otherwise).

| Model | Fits | Median range | Largest range | Range above 1 | Selected best | Selected worst |
|---|---|---|---|---|---|---|
| P | 26 | 0.04 | 2.34 | 1 | 6 | 9 |
| V-pen | 26 | 0.11 | 1.46 | 1 | 5 | 5 |
| V-none | 26 | 0.13 | 1.76 | 2 | 10 | 7 |
| A-hard | 26 | 0.01 | 0.28 | 0 | 12 | 5 |
| A-pen | 26 | 0.03 | 0.54 | 0 | 7 | 4 |
| A-none | 26 | 0.04 | 1.41 | 2 | 6 | 4 |
| C1 | 26 | 0.01 | 2.84 | 1 | 4 | 2 |
| F-hard | 26 | 0.54 | 1.48 | 5 | 7 | 7 |
| F-pen | 26 | 0.67 | 1.28 | 3 | 4 | 2 |
| F-none | 26 | 0.26 | 0.95 | 0 | 8 | 5 |
| F-rel | 26 | 0.18 | 0.97 | 0 | 3 | 10 |
| P-K | 2 | 0.34 | 0.47 | 0 | 0 | 1 |

Table S4. Seed spread of the held-out error over the E-high and E-low fits. Range: the difference between the largest and the smallest $\log_{10} E$ over the 5 seeds of one fit, in decades; Range above 1: fits whose seeds differ by more than a factor of ten. Selected best (worst): fits in which the seed with the lowest training loss, the one reported, had the best (worst) held-out error of the five.

| Truth | Comparator $X$ | $R$ | 95 % interval | Holm $p$ | Failed P/$X$ | Verdict |
|---|---|---|---|---|---|---|
| KS | V-pen | 0.827 | 0.74–0.978 | 0.024 | 0/0 | differ, size uncertain |
| | A-hard | 0.278 | 0.155–0.42 | 0.004 | 0/0 | P better |
| | F-none | 0.0259 | 0.0102–0.0785 | 0.004 | 0/5 | P better |
| DA | V-pen | 0.982 | 0.766–1.21 | 1.000 | 0/0 | equivalent |
| | A-hard | 114 | 72.4–181 | 0.004 | 0/0 | A-hard better |
| | F-none | 0.0561 | 0.0187–0.163 | 0.004 | 0/2 | P better |
| FG | V-pen | 1.27 | 0.266–1.7 | 1.000 | 0/3 | inconclusive |
| | A-hard | 0.41 | 0.363–0.545 | 0.004 | 0/0 | P better |
| | F-none | 0.0322 | 0.00624–0.122 | 0.004 | 0/6 | P better |
| HS | V-pen | 0.658 | 0.528–0.837 | 0.017 | 0/0 | P better |
| | A-hard | 0.22 | 0.0913–0.423 | 0.004 | 0/0 | P better |
| | F-none | 0.0269 | 0.00575–0.0843 | 0.004 | 0/5 | P better |
| TC | V-pen | 0.911 | 0.67–1.17 | 1.000 | 0/0 | inconclusive |
| | A-hard | 0.023 | 0.0124–0.0436 | 0.004 | 0/0 | P better |
| | F-none | 0.00687 | 0.00198–0.026 | 0.004 | 0/4 | P better |

Table S5. The fifteen isosteric-heat tests on the synthetic truths (Note S2). $R$: Hodges-Lehmann ratio of the root-mean-square error of $Q_{\mathrm{st}}$ at matched loading, P over $X$, over the 13 materials, with its exact 95% interval; Holm $p$ over the fifteen tests; Failed: materials on which a task failed (an undefined or non-finite heat, too few states in the loading window), ranked worst.

| Dataset | $n$ | H1b V-pen | H1c A-hard | H1d F-none | H1e C1 |
|---|---|---|---|---|---|
| confirmatory set | 13 | 1.10 [0.93, 1.65] inc | 1.51 [0.84, 2.87] inc | 0.63 [0.40, 1.22] inc | 1.23 [0.92, 1.96] inc |
| coverage rule, first version | 6 | 0.97 [0.83, 1.27] eq | 0.88 [0.32, 1.48] inc | 0.46 [0.32, 1.61] inc | 1.11 [0.74, 3.20] inc |
| coverage rule, second version | 9 | 1.19 [0.94, 3.32] inc | 1.76 [0.73, 3.92] inc | 1.00 [0.49, 2.48] inc | 1.49 [0.95, 2.95] inc |
| no coverage rule | 23 | 1.05 [0.97, 1.20] eq | 1.52 [1.06, 2.19] inc | 0.59 [0.42, 0.79] P | 1.16 [0.97, 1.58] inc |
| 7 points per isotherm | 14 | 1.05 [0.89, 1.54] inc | 1.29 [0.73, 2.67] inc | 0.63 [0.41, 1.04] inc | 1.20 [0.91, 1.66] inc |
| $Co_2Cl_2$(BTDD) from its characteristic curves | 13 | 1.04 [0.93, 1.58] inc | 1.51 [0.84, 2.87] inc | 0.69 [0.40, 1.32] inc | 1.28 [0.96, 2.03] inc |
| digitisation shift, copy 1 | 13 | 1.02 [0.90, 1.77] inc | 1.35 [0.86, 3.06] inc | 0.58 [0.38, 1.07] inc | 1.35 [0.94, 2.22] inc |
| digitisation shift, copy 2 | 13 | 1.15 [0.96, 1.72] inc | 1.34 [0.72, 2.29] inc | 0.53 [0.37, 0.78] P | 1.35 [0.97, 1.82] inc |

Table S6. The H1 comparisons on the pre-registered dataset variants: $R$ with its exact 95% interval and the verdict (P, P better; X, comparator better; eq, equivalent within 25%; inc, inconclusive). The coverage variants are earlier versions of the coverage rule and the dataset without it; the digitisation shifts move every digitised isotherm by $\pm 0.002$ in $p/p_{\mathrm{sat}}$.

| Comparison | $X$ | $R$ | Bootstrap 95 % | $R$, MOFs ($n = 5$) | $R$, COFs ($n = 8$) |
|---|---|---|---|---|---|
| H1b | V-pen | 1.10 | 0.94–1.58 | 1.09 | 1.10 |
| H1c | A-hard | 1.51 | 0.87–2.67 | 0.69 | 2.67 |
| H1d | F-none | 0.63 | 0.42–1.07 | 0.43 | 0.93 |
| H1e | C1 | 1.23 | 0.94–1.91 | 0.99 | 1.47 |
| H1a (descriptive) | F-hard | 0.49 | 0.33–0.82 | 0.48 | 0.50 |
| H1b by source paper | V-pen | 1.12 | exact 0.92–3.08 | 10 sources | |

Table S7. The H1 comparisons with the percentile bootstrap interval (pre-registered as a secondary), split into MOFs and COFs (pre-registered subgroup, Hodges-Lehmann estimates only, no intervals at these sizes), and, as an exploratory check of the dependence between materials from one paper, H1b with the per-material differences averaged within each source paper (ledger X-4).

## Note S1: Can each verdict fail? Synthetic worlds and power

Before the freeze, the whole pipeline (the tuned configurations, fitting from 5 seeds, the violation counts and the analysis code) was run in synthetic worlds that share no uptake data with the confirmatory set. Each world has 13 materials; material $i$ uses the temperatures of confirmatory material $i$. An expected outcome was written for each world before it ran. In KS0 the truth is a two-site Sips isotherm with van't Hoff affinities, inside the family of P; in DA it is a Dubinin-Astakhov curve, inside the family of the hard characteristic curve (A-hard); in FW it is a Kirchhoff-Sips isotherm with a large heat capacity, so that van't Hoff is wrong; FLAT has one weakly curved Langmuir site; SUB has a sub-linear Sips isotherm outside the family of P and inside that of the penalty model V-pen.

The comparison of P with A-hard discriminated in both directions: P extrapolated better in KS0 ($R = 0.52$, exact 95% interval 0.35 to 0.70) and A-hard better in DA ($R = 1.26$, 1.07 to 1.49). Three expectations were missed. In FW the comparison of P with the free temperature network F-hard still read "P better" ($R = 0.33$): with two training temperatures the free network cannot extrapolate, so this comparison cannot fail and is reported descriptively. In FLAT the unconstrained network F-none violated physics "often". In SUB, built for V-pen to win, P was slightly better ($R = 0.96$, 0.94 to 0.99); the mixture of Langmuir terms in P approximates a sub-linear isotherm over the measured range. The accuracy comparison of hard constraints with penalties therefore could not be made to fail synthetically, and it is tested as non-inferiority. The rival claim, that penalties suffice for accuracy, predicts the same equivalence, so accuracy alone cannot separate the two approaches; admissibility (H3) is the comparison that can.

The power of each verdict was simulated with the frozen analysis code at $n = 13$ (2000 simulations per setting). At the spread expected for the comparison with V-pen (0.158 decades), the non-inferiority criterion is met with probability 0.0225 when the true ratio is at the margin, $R = 1.25$, and 0.51 (normal) or 0.74 (Student $t$, 3 degrees of freedom) when the two models are equally accurate. A "not met" outcome is therefore likely even under equal accuracy and is never read as evidence that the constraints cost accuracy.

## Note S2: Synthetic truths for the isosteric heat

The heat test uses five truth families with exact heats. Their shape parameters are fitted by bounded least squares to each confirmatory material's middle-temperature isotherm; their thermodynamic parameters are fixed in advance: KS, a Langmuir site with $Q_0 = 55$ kJ mol$^{-1}$ plus a cooperative Sips site with $Q_0 = 48$ kJ mol$^{-1}$, a heat capacity of 40 J mol$^{-1}$ K$^{-1}$ and an exponent of at least 1.5, inside the family of the Kirchhoff variant of P and near that of P; DA, a Dubinin-Astakhov curve $q = W_0 \exp[-(A/E)^n]$ with $1 \le n \le 6$, inside the family of the characteristic curve; FG, a Fowler-Guggenheim isotherm with $Q_0 = 45$ kJ mol$^{-1}$ and a lateral attraction between 0 and $3.9\,RT_{\min}$, inside neither; HS, strong hydrophilic sites, a Langmuir site with $Q = 80$ kJ mol$^{-1}$ holding at most 0.3 $q_{\mathrm{ref}}$ plus a Sips site with $Q = 46$ kJ mol$^{-1}$, inside the family of P with a heat that falls steeply with loading; TC, a Sips isotherm with $Q = 50$ kJ mol$^{-1}$ whose capacity falls by 0.2% per kelvin, inside the family of P and outside that of the characteristic curve. Gaussian noise of 0.4% of $q_{\mathrm{ref}}$ is added with a fixed seed per material and family.

## Note S3: Digitisation error

The digitiser was tested against published figures whose numeric data are also published, before any model was fitted (logs/digitisation_error.json, logs/digitisation_validation.json). On the vector figure of Mg-SU-102[1] the extraction from the PDF drawing commands recovered 198 of 198 points with a root-mean-square uptake error of $7 \times 10^{-6}$ g g$^{-1}$ and a largest error of

$5 \times 10^{-5}$ g g$^{-1}$. The same figure rasterised at 150, 300 and 600 dpi and read by template matching gave root-mean-square uptake errors of $6.3 \times 10^{-4}$, $4.0 \times 10^{-4}$ and $3.5 \times 10^{-4}$ g g$^{-1}$, with no false detection; markers drawn hollow are not read by the raster method. On raster figures of HKUST-1 from another paper[2], the root-mean-square uptake error per series was 0.11 to 0.19% of the axis span, or 0.0012 to 0.0034 of the series' largest uptake, again with no false detection; markers hidden under other markers are missed (recall of the visible markers 0.91 to 1.00). On vector figures of two further papers[3, 4] read through their legend colours, every point was recovered and the root-mean-square uptake error was 0.04 to 0.08% of the axis span. The noise level of the synthetic tests, 0.4% of $q_{\mathrm{ref}}$, is the largest raster error per isotherm maximum rounded up.

## Note S4: Model definitions

All models map $s = \ln(p/\mathrm{Pa})$ and $T$ to the uptake $q$ (mol kg$^{-1}$); the code is src/p3a/models.py, and the tuned settings and parameter counts are in Table S8. $T_{\mathrm{ref}} = 300$ K throughout. The same symbol $q$ is used for the uptake of every model; Figure 1a of the main text writes $q^*$ for P's uptake as a derivative of $\psi$.

**P (PotentialIsotherm, van't Hoff).** The reduced spreading pressure is

$$\psi(s,T) = \sum_k \frac{c_k(T)}{g_k}\,\mathrm{softplus}\big(g_k(s-\theta_k(T))\big), \qquad q = \frac{\partial\psi}{\partial s} = \sum_k c_k(T)\,\sigma\big(g_k(s-\theta_k(T))\big),$$

with $\sigma$ the logistic function. Each term is therefore a Langmuir term ($g_k = 1$, the $n_{\mathrm{H}}$ Henry terms) or a cooperative Sips-type term ($g_k = 1.5 + \mathrm{softplus}(\cdot)$, the $n_{\mathrm{c}}$ cooperative terms) in $K_k(T)p$, with $\ln K_k = -\theta_k$. The capacities are $c_k(T) = c_{k0}\exp[-\alpha_k(T - T_{\mathrm{ref}})]$ with $c_{k0} > 0$ (a softplus), and the positions follow van't Hoff, $\theta_k(T) = \theta_{k0} - (Q_k/R)(1/T - 1/T_{\mathrm{ref}})$. The heats $Q_k$ and the expansion coefficients $\alpha_k$ are unconstrained in sign. Positive capacities make $q$ strictly increasing in $s$; the Henry terms give $q/p \to \sum_{\mathrm{Henry}} c_k K_k > 0$ as $p \to 0$; the heat $Q_{\mathrm{st}} = -RT^2(\partial q/\partial T)/(\partial q/\partial s)$ has a positive denominator and is finite everywhere, but its sign and size are not constrained. P is thus a positive mixture of Langmuir and cooperative terms with van't Hoff affinities, written as the derivative of one convex potential; C1 is a two-term member of the same broad family with free exponents.

**V-none and V-pen.** The same function, the same initial function for a given seed and the same tuned structure as P, but with signed capacities and every exponent free (including those of the Henry terms). V-pen adds the penalties below with weight $\lambda$.

**P-K.** As P, with a constant heat capacity $\Delta c_k$ per term: $\theta_k(T) = \theta_{k0} - (Q_k/R)(1/T - 1/T_{\mathrm{ref}}) - (\Delta c_k/R)[\ln(T/T_{\mathrm{ref}}) + T_{\mathrm{ref}}/T - 1]$.

**F-hard.** The same potential and shape constraints as P, but $c_k(T)$ and $\theta_k(T)$ are outputs of a one-hidden-layer tanh network of the scaled inverse temperature $\hat{\tau} = (T_{\mathrm{ref}}/T - 1)/0.1$, with each $\theta_k$ kept within a bounded range around the data.

**A-hard, A-none, A-pen.** The characteristic curve $q = W(A)$, $A = RT\ln(p_{\mathrm{sat}}(T)/p)$, with $W(A) = \sum_k c_k\,\sigma(-g_k(A/A_{\mathrm{s}} - A_k))$, $A_{\mathrm{s}} = 2$ kJ mol$^{-1}$. In A-hard, $c_k > 0$ and $g_k > 0$, so $W$ decreases with $A$ and $q$ increases with $p$ at every temperature; the model is temperature-invariant in $A$ by construction, has no Henry limit, and its heat is exactly $\Delta_{\mathrm{vap}}H(T) + A$. A-none has signed $c_k$ and free $g_k$; A-pen adds the monotonicity penalty.

**F-none, F-pen, F-rel.** A fully connected tanh network with inputs $((s - s_c)/s_w, \hat{\tau})$ and one linear output scaled by the largest training uptake (F-none, F-pen), or with $\ln(p/p_{sat})$ in place of $s$ (F-rel); width and depth in Table S8.

**C1.** $q = \sum_{i=1}^{2} q_{m,i}\,\sigma(g_i(s + \ln b_i(T)))$, that is $q_{m,i}(b_i p)^{g_i}/[1 + (b_i p)^{g_i}]$, with $q_{m,i} > 0$, $g_i = 0.2 + \mathrm{softplus}(\cdot)$ and $\ln b_i(T) = \ln b_{i,\mathrm{ref}} + (Q_i/R)(1/T - 1/T_{\mathrm{ref}})$.

**P-sign (exploratory; ledger X-6).** P with $Q_k = \mathrm{softplus}(\cdot) > 0$ and $\alpha_k = 10^{-3}\,\mathrm{softplus}(\cdot) \geq 0\ \mathrm{K}^{-1}$, everything else unchanged; the constructor draws the same random numbers as P, so a seed gives the same initial function (checked: maximum difference zero). With both signs fixed, every term's affinity and capacity decrease with temperature, so the heat is positive by construction. The variant was designed after the confirmatory results and fitted on all three tasks (scripts/explore_p_signs.py).

**Initial values and data handling.** The heats start at $Q_k = 50\ \mathrm{kJ\,mol^{-1}}$ plus a seeded jitter of standard deviation $5\ \mathrm{kJ\,mol^{-1}}$, the expansion coefficients at zero, and the term positions at evenly spaced offsets around the centre of the training pressures. Saturation pressures are from the IAPWS equation[5]. Uptakes were converted to $\mathrm{mol\,kg^{-1}}$ with $M = 18.01528\ \mathrm{g\,mol^{-1}}$ ($\mathrm{g\,g^{-1}}$, wt %) and $22.414\ \mathrm{L\,mol^{-1}}$ ($\mathrm{cm^3(STP)\,g^{-1}}$). In I-mid with an even number of temperatures, the lower of the two middle temperatures is held out.

**Penalties.** On a collocation grid of 60 pressures (1 Pa to $0.9\,p_{sat}$) at 9 temperatures from $T_{min} - 15$ K to $T_{max} + 15$ K, the monotonicity penalty is the mean of $\max(0, -\partial q/\partial s)^2$ divided by the squared uptake scale. The Henry penalty, at 0.01 Pa, is the mean of $[(\partial q/\partial s - q)/(|\partial q/\partial s| + |q|)]^2$ plus the same relative measure of negative uptake; it is zero only for an isotherm linear in $p$. The characteristic-curve models have no Henry penalty.

**Fitting.** The loss is the mean squared error of $q$ divided by the largest training uptake, plus $\lambda$ times the penalty and an optional $\ell_2$ term; Adam (learning rate $10^{-2}$) for the number of steps in Table S8, then L-BFGS for 300 iterations, in double precision from 5 seeds, keeping the seed with the lowest training loss. Inputs are centred on the training mean of $s$ and scaled by its standard deviation (at least 0.5).

| Model | Role | Form | Tuned settings | Parameters |
|---|---|---|---|---|
| P | confirmatory reference | potential, van't Hoff | $n_{\mathrm{H}}$ 2, $n_{\mathrm{c}}$ 2, Adam steps 2000 | 18 |
| V-pen | H1b (primary) | as P, constraints as penalty | $n_{\mathrm{H}}$ 2, $n_{\mathrm{c}}$ 2, $\lambda$ 0.001, Adam steps 2000 | 20 |
| V-none | descriptive | as P, unconstrained | $n_{\mathrm{H}}$ 2, $n_{\mathrm{c}}$ 2, Adam steps 2000 | 20 |
| A-hard | H1c | characteristic curve | $n$ 6, Adam steps 4000 | 18 |
| A-pen | H3 only | as A-hard, penalty | $n$ 6, $\lambda$ 0.001, Adam steps 4000 | 18 |
| A-none | H3 only | as A-hard, unconstrained | $n$ 6, Adam steps 4000 | 18 |
| F-none | H1d | network in $(s, 1/T)$ | width 32, depth 3, $\ell_2$ 0.0001, Adam steps 4000 | 2241 |
| F-pen | H3 only | as F-none, penalty | width 32, depth 3, $\ell_2$ 0.0001, $\lambda$ 0.1, Adam steps 4000 | 2241 |
| F-hard | H1a (descriptive) | potential, network in $1/T$ | $n_{\mathrm{H}}$ 1, $n_{\mathrm{c}}$ 2, width 8, Adam steps 2000 | 72 |
| C1 | H1e | dual-site Sips, van't Hoff | Adam steps 4000 | 8 |
| F-rel | descriptive | network in $(\ln p/p_{\mathrm{sat}}, 1/T)$ | width 32, depth 3, Adam steps 2000 | 2241 |
| P-K | descriptive | P with heat capacity | $n_{\mathrm{H}}$ 2, $n_{\mathrm{c}}$ 2, Adam steps 2000 | 22 |

Table S8. The models, their role in the analysis, the settings chosen on the development set (Methods), and the number of fitted parameters. $n_{\mathrm{H}}$, $n_{\mathrm{c}}$: Henry and cooperative terms; $n$: terms of the characteristic curve; width and depth of a network; $\ell_2$: weight decay; $\lambda$: penalty weight. Every fit then ran L-BFGS for 300 iterations.

## Table S9: Penalty weight (exploratory, after the verdicts)

| Weight / $\lambda$ | $R$ | 95 % interval | $p$ | P better | In domain | Band | Strict | Penalty value |
|---|---|---|---|---|---|---|---|---|
| 1 (tuned) | 1.10 | 0.93–1.65 | 0.38 | 6/13 | 9 | 10 | 11 | – |
| $10^2$ | 1.07 | 0.91–1.70 | 0.59 | 6/13 | 4 | 4 | 6 | 0 |
| $10^4$ | 1.09 | 0.94–1.84 | 0.59 | 7/13 | 2 | 2 | 4 | 0 |
| $10^6$ | 1.12 | 0.96–1.64 | 0.50 | 6/13 | 0 | 0 | 5 | 0 |

Table S9. V-pen refitted on the 13 confirmatory materials (E-high and E-low) with the monotonicity penalty on its own weight, a multiple of the tuned $\lambda = 0.001$, and the Henry-slope penalty kept at $\lambda$; all other settings, the collocation points and the 5 seeds as in the confirmatory run (ledger X-7; exploratory, specified after the verdicts). The first row is the confirmatory V-pen. $R$: Hodges-Lehmann ratio of P's error over V-pen's with its exact interval, from the frozen analysis code; $p$: exact Wilcoxon, unadjusted, no verdict class; P better: materials on which P had the lower error. In domain, Band and Strict: violating fits out of 26, defined as in Table S3. Penalty value: median over the fits of the monotonicity penalty at the reported seed, on its collocation points.

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